

# Physics of Semiconductor Devices

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ECE210: PSD



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DELHI

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week 02



# Classical Mechanics and Quantum Mechanics



**Mechanics:** the study of the behavior of physical bodies when subjected to forces or displacements



**Classical Mechanics:** describing the motion of **macroscopic** objects.

**Macroscopic:** measurable or observable by naked eyes

**Quantum Mechanics:** describing behavior of systems at atomic length scales and smaller.

# Wave-Particle Duality



Particle-like wave behavior  
(example, photoelectric effect)

Wave-like particle behavior  
(example, Davisson-Germer experiment)

Wave-particle duality

## Mathematical descriptions:

The momentum of a photon is:  $p = \frac{h}{\lambda}$

The wavelength of a particle is:  $\lambda = \frac{h}{p}$

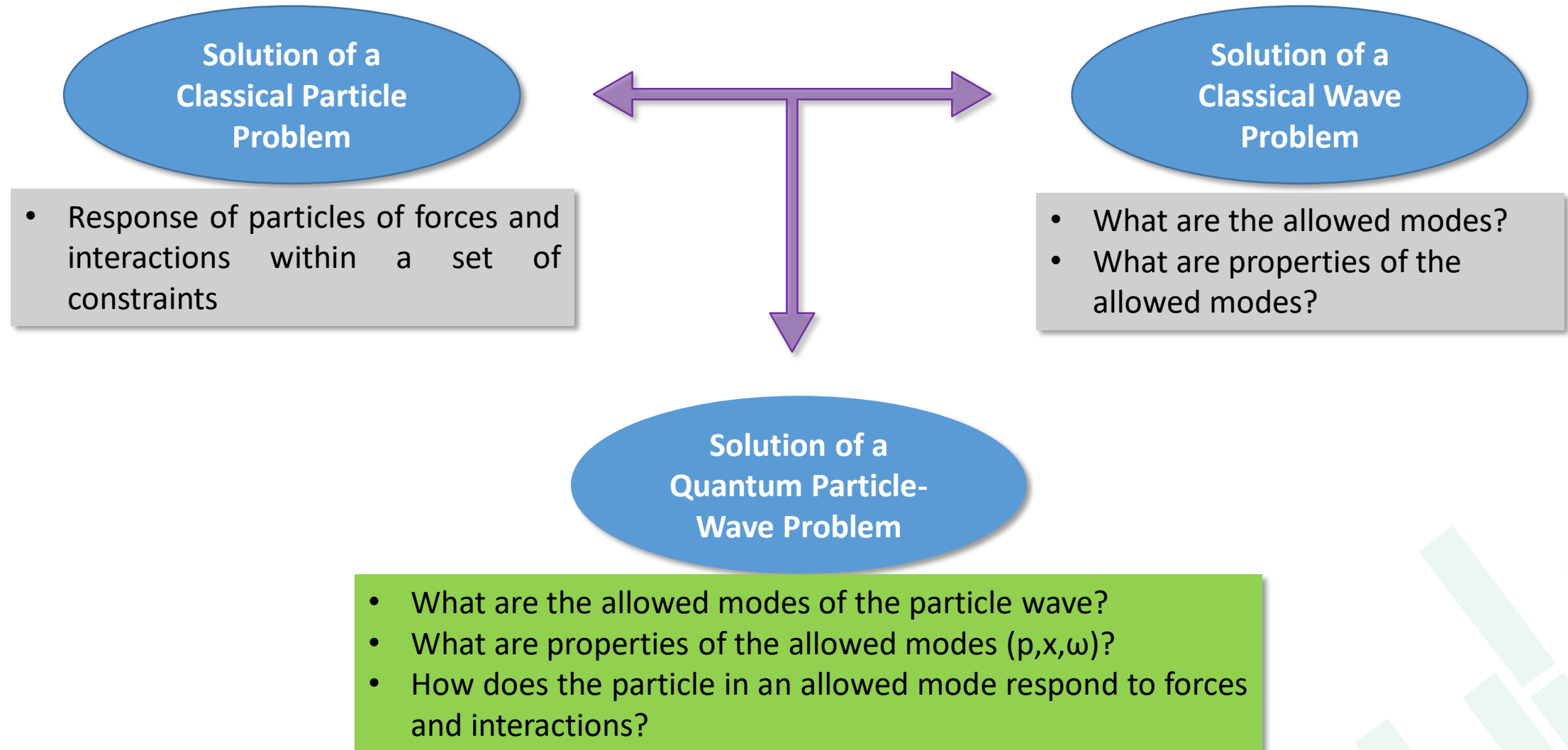
$\lambda$  is called the *de Broglie wavelength*

## When a object that appears as particle can behave as wave?

- $\lambda_{\text{BASEBALL}} \approx 10^{-34}\text{m}$  ( $m = 0.15 \text{ Kg}$ ,  $v = 40\text{m/s}$ )
- $\lambda_{\text{ELECTRON}} \approx 10^{-10}\text{m}$  ( $m = 9.11 \times 10^{-31} \text{ Kg}$ ,  $v = 10^6\text{m/s}$ )

When the dimension ( $r$ ) over which the change of potential energy  $V(r)$  of a particle becomes smaller/comparable to its wavelength, its wave nature reveals.

# Foundations of Quantum Mechanics



# The Uncertainty Principle



## The Heisenberg Uncertainty Principle (year 1927):

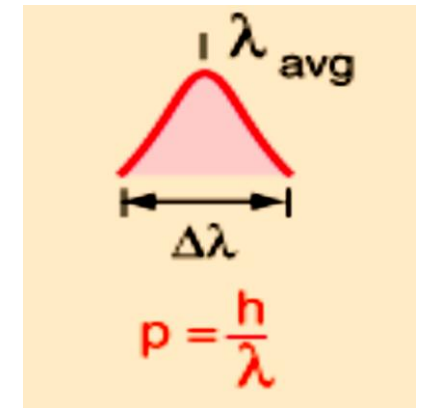
- It is impossible to simultaneously describe with absolute accuracy the position and momentum of a particle

$$\Delta p \Delta x \geq \hbar$$

- It is impossible to simultaneously describe with absolute accuracy the energy of a particle and the instant of time the particle has this energy

$$\Delta E \Delta t \geq \hbar$$

The Heisenberg uncertainty principle applies to electrons and states that we can not determine the exact position of an electron. Instead, we could determine the probability of finding an electron at a particular position.



A continuous distribution of wavelengths can produce a localized wave packet.

**Zero-Point Energy** (vibrational energy  $E_{\text{vib}} \neq 0$ , even at  $T = 0$  K) is also a consequence of the Uncertainty Principle. If vibration ceases at  $T = 0$ , then position and momentum both = 0 (violating the UP).

# Wavefunction

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- Quantum mechanics acknowledges the wave-particle duality of matter by supposing that, rather than traveling along a definite path, a particle is distributed through space like a wave. The wave that in quantum mechanics replaces the classical concept of particle trajectory is called a wavefunction,  $\psi$  (“psi”).

A **wave function** in [quantum mechanics](#) describes the [quantum state](#) of an **isolated** system of one or more particles. There is *one* wave function containing all the information about the entire system.

# Postulates of Quantum Mechanics



## Postulate 1:

### State and wave functions. Born interpretation

The state of a quantum mechanical system is completely specified by a wave function  $\psi(\mathbf{r}, t)$  that depends on the coordinates of the particles ( $\mathbf{r}$ ) and time  $t$ . These functions are called wave functions or state functions.

For 2 particle system:  $\Psi = \Psi(x_1, y_1, z_1, x_2, y_2, z_2, t)$

Wave function contains all the information about a system.

|                     |                   |                       |
|---------------------|-------------------|-----------------------|
| wave function       | $\Leftrightarrow$ | classical trajectory  |
| (Quantum mechanics) |                   | (Newtonian mechanics) |

**Meaning** of wave function:

$$P(\mathbf{r}) = |\psi|^2 = \int \psi^* \psi d\tau$$

=> the probability that the particle can be found at a particular point  $\mathbf{x}$  and a particular time  $t$ . (Born's / Copenhagen interpretation)

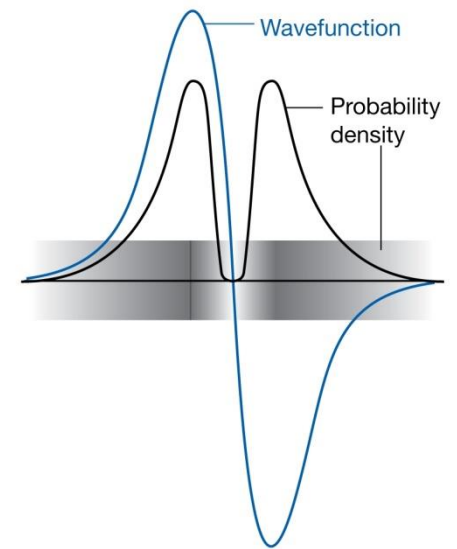
# Implications of Born's Interpretation



(1) Positivity:

$$P(\mathbf{r}) \geq 0$$

The sign of a wavefunction has no direct physical significance:  
The positive and negative regions of this wavefunction both correspond to the same probability distribution.



(2) Normalization: 
$$\int_{all\_space} \Psi^* \Psi d\tau = 1$$

i.e. the probability of finding the particle in the universe is 1.

# Contd...



## Postulate 2

To every physical property, observable in classical mechanics, there corresponds a linear, Hermitian operator in quantum mechanics.

**Classical-Mechanical Observables and Their Corresponding Quantum-Mechanical Operators**

| Observable       |                        | Operator                                                   |                                                                                                                                                                                                             |
|------------------|------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name             | Symbol                 | Symbol                                                     | Operation                                                                                                                                                                                                   |
| Position         | $x$                    | $\hat{X}$                                                  | Multiply by $x$                                                                                                                                                                                             |
|                  | $\mathbf{r}$           | $\hat{\mathbf{R}}$                                         | Multiply by $\mathbf{r}$                                                                                                                                                                                    |
| Momentum         | $p_x$                  | $\hat{P}_x$                                                | $-i\hbar \frac{\partial}{\partial x}$                                                                                                                                                                       |
|                  | $\mathbf{p}$           | $\hat{\mathbf{P}}$                                         | $-i\hbar(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z})$                                                                            |
| Kinetic energy   | $T_x$                  | $\hat{T}_x$                                                | $-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$                                                                                                                                                       |
|                  | $T$                    | $\hat{T}$                                                  | $-\frac{\hbar^2}{2m}(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2})$<br>$= -\frac{\hbar^2}{2m} \nabla^2$                                              |
| Potential energy | $V(x)$<br>$V(x, y, z)$ | $\hat{V}(\hat{x})$<br>$\hat{V}(\hat{x}, \hat{y}, \hat{z})$ | Multiply by $V(x)$<br>Multiply by $V(x, y, z)$                                                                                                                                                              |
| Total energy     | $E$                    | $\hat{H}$                                                  | $-\frac{\hbar^2}{2m} \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right)$<br>$+ V(x, y, z)$<br>$= -\frac{\hbar^2}{2m} \nabla^2 + V(x, y, z)$ |
| Angular momentum | $L_x = yp_z - zp_y$    | $\hat{L}_x$                                                | $-i\hbar \left( y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right)$                                                                                                                      |
|                  | $L_y = zp_x - xp_z$    | $\hat{L}_y$                                                | $-i\hbar \left( z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right)$                                                                                                                      |
|                  | $L_z = xp_y - yp_x$    | $\hat{L}_z$                                                | $-i\hbar \left( x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right)$                                                                                                                      |

## Postulate 3

In any measurement of the observable associated with the operator  $\hat{A}$ , the only values that will ever be observed are the eigenvalues 'a' which satisfy the eigenvalue equation:

$$\hat{A}\Psi = a\Psi$$

This is the postulate that the values of dynamical variables are quantized in quantum mechanics.

### Eigen Function and Eigen value

$$\hat{A} f(x) = k f(x)$$

$f(x)$  is eigenfunction of  $\hat{A}$  with eigenvalue  $k$

### Eigenvalue equation

Eigenvalue equation

(Operator)(function) = (constant factor)\*(same function)

Example:

$e^{ikx}$  is an eigenfunction of a operator  $\hat{P}_x = -i\hbar \frac{\partial}{\partial x}$

$$F(x) = e^{ikx} \Rightarrow -i\hbar \frac{\partial}{\partial x} e^{ikx} = -i^2 \hbar k^2 e^{ikx} = \hbar k^2 e^{ikx}$$

Thus  $e^{ikx}$  is an eigenfunction

# Contd...



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|                  | $L_z = xp_y - yp_x$    | $\hat{L}_z$                                                | $-i\hbar \left( x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right)$                                                                                                                      |

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# Significance of commutation rules

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- The eigenvalue of a Hermitian operator is real.
- A real eigenvalue means that the physical quantity for which the operator stands for can be measured experimentally.
- The eigenvalues of two commuting operators can be computed by using the common set of eigenfunctions.

**If the two operators commute, then it is possible to measure the simultaneously the precise value of both the physical quantities for which the operators stand for.**

Question: Find commutator of the operators  $x$  and  $p_x$

Is it expected to be a non-zero or zero quantity?

Hint: Heisenberg Uncertainty Principle



# Contd...



Commute or not ??

$$\hat{A} = \hat{x}$$

- Operator  $x$  and  $d/dx$

$$\hat{B} = \frac{d}{dx}$$

They don't

$$\hat{A} \hat{B} f(x) = x f'(x)$$

$$\hat{B} \hat{A} f(x) = \frac{d}{dx} [x f(x)] = x f'(x) + f(x)$$

$$[\hat{A}, \hat{B}] f(x) = -f(x)$$

$$[\hat{A}, \hat{B}] = -1$$

## Postulate 4

For a system in a state described by a normalized wave function  $\Psi$ , the average or expectation value of the observable corresponding to  $A$  is given by:

$$\langle A \rangle = \int \Psi^* \hat{A} \Psi d\tau$$

Expectation value in general:

$$\langle a \rangle = \frac{\int_{-\infty}^{\infty} \Psi^*(x) \hat{A} \Psi(x) dx}{\int_{-\infty}^{\infty} \Psi^*(x) \Psi(x) dx} = \int_{-\infty}^{\infty} \Psi^*(x) \hat{A} \Psi(x) dx \quad (\text{if normalized})$$

The fourth postulates states what will be measured when large number of identical systems are interrogated one time. Only after large number of measurements it will converge to  $\langle a \rangle$ .

In QM, the act of the measurement causes the system to “collapse” into a single eigenstate and in the absence of an external perturbation it will remain in that eigenstate.

## Postulate 5

The wave function of a system evolves in time in accordance with the time dependent Schrodinger equation:

$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H} \Psi$$



# Quantum Theory for Semiconductors



How to determine the behavior of electrons and holes in the semiconductor?

- Mathematical description of motion of electrons in quantum mechanics — **Schrödinger's Wave Equation**

The Schrodinger equation plays the role of Newton's laws and conservation of energy in classical mechanics - i.e., it predicts the future behavior of a dynamic system.

It is a wave equation in terms of the wave function which predicts analytically and precisely the probability of events or outcome

- Solution of Schrödinger's Wave Equation: energy band structure and probability of finding a electron at a particular position

# Schrödinger's Wave Equation



One dimensional **Schrödinger's Wave Equation**:

$$\frac{-\hbar^2}{2m} \frac{\partial^2 \Psi(x,t)}{\partial x^2} + V(x)\Psi(x,t) = j\hbar \frac{\partial \Psi(x,t)}{\partial t}$$

$$\hat{\mathbf{H}}\psi = E\psi$$



Hamiltonian

where

$$\hat{\mathbf{H}} = \hat{\mathbf{T}} + \hat{\mathbf{V}}$$

Energy eigenvalue

$$\hat{\mathbf{T}} = \frac{\hat{\mathbf{p}}^2}{2m} = -\frac{\hbar^2}{2m} \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) = -\frac{\hbar^2}{2m} \nabla^2$$

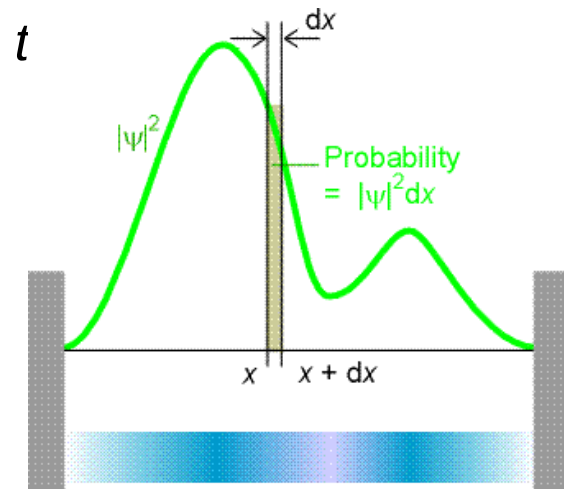
$\Psi(x,t)$  : Wave function

$|\Psi(x,t)|^2 dx$  , the **probability** to find a particle in  $(x, x+dx)$  at time  $t$

$|\Psi(x,t)|^2$  , the **probability density** at location  $x$  and time  $t$

$V(x)$  : Potential function

$m$  : Mass of the particle



# 3-D System:

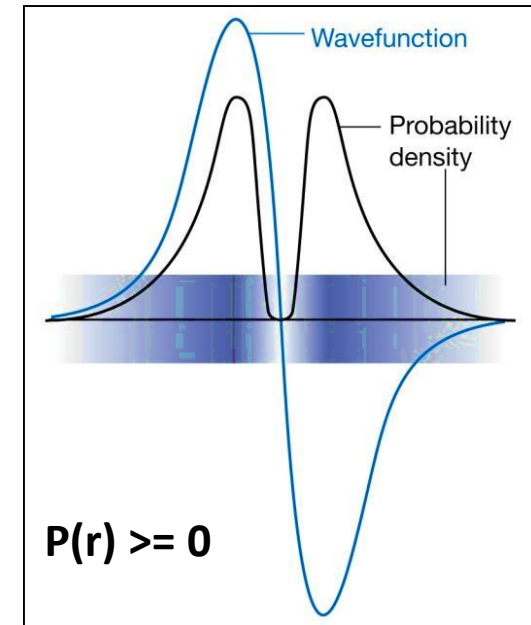


- If the wavefunction at  $\mathbf{r} = (x,y,z)$  is  $\psi(\mathbf{r})$ , the probability of finding the particle in the infinitesimal volume element  $d\tau$  ( $= dx dy dz$ ) is:

$$P(\mathbf{r}) \propto |\psi(\mathbf{r})|^2 d\tau$$

- If  $\psi(\mathbf{r})$  is the wavefunction describing the spatial distribution of an electron in an atom or molecule, then:

$$|\psi(\mathbf{r})|^2 = \rho(\mathbf{r}) \text{ — the electron density at point } \mathbf{r}$$



SE can be set up for any physical system. The form of  $\hat{H}$  depends on the system.

The Schrodinger equation gives the quantized energies of the system and gives the form of the wavefunction so that other properties may be calculated

# Physically acceptable wave function



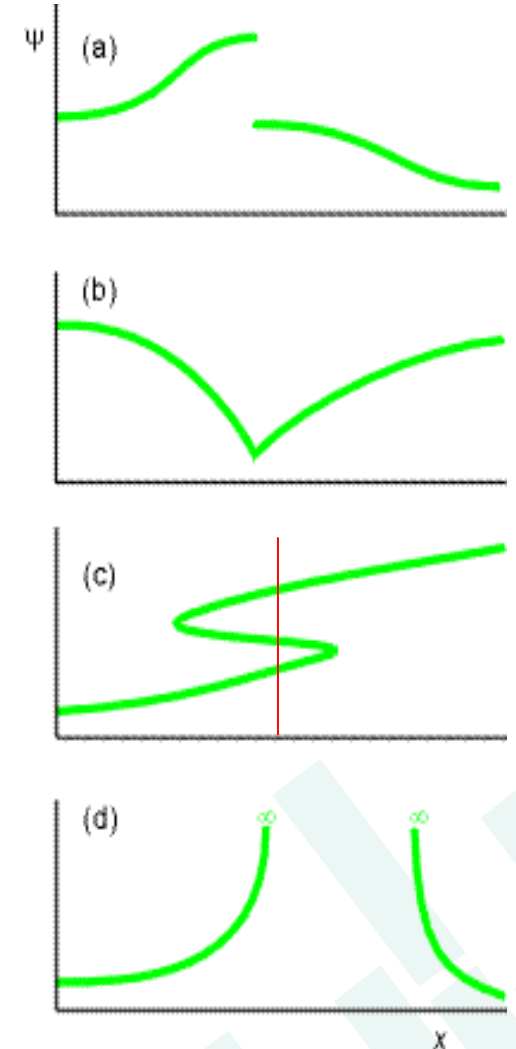
The Born interpretation of  $\Psi$  places restrictions on the form of the wavefunction:

- (a)  $\Psi$  must be continuous (no breaks);
- (b) The gradient of  $\Psi$  ( $d\Psi/dx$ ) must be continuous (no kinks);
- (c)  $\Psi$  must have a single value at any point in space;
- (d)  $\Psi$  must be finite everywhere;
- (e)  $\Psi$  cannot be zero everywhere.

## Quantization of the Wavefunction:

- Other restrictions (boundary conditions) depend on the exact system.
- These restrictions on  $\psi$  mean that only certain wavefunctions and only certain energies of the system are allowed.

$\therefore$  Quantization of  $\psi \Rightarrow$  Quantization of  $E$



**The wave function must be single valued.** This means that for any given values of  $x$  and  $t$ ,  $\Psi(x,t)$  must have a unique value. This is a way of guaranteeing that there is only a single value for the probability of the system being in a given state.

**The wave function must be square-integrable:** In other words, the integral of  $|\Psi|^2$  over all space must be finite. This is another way of saying that it must be possible to use  $|\Psi|^2$  as a probability density, since any probability density must integrate over all space to give a value of 1, which is clearly not possible if the integral of  $|\Psi|^2$  is infinite. One consequence of this proposal is that  $|\Psi|^2$  must tend to 0 for infinite distances.

**Continuous wavefunction:** A rapid change would mean that the derivative of the function was very large (either a very large positive or negative number). In the limit of a step function, this would imply an infinite derivative. Since the momentum of the system is found using the momentum operator, which is a first order derivative, this would imply an infinite momentum, which is not possible in a physically realistic system.

**Continuous First derivative:** All first-order derivatives of the wave function must be continuous. Following the same reasoning as in condition 3, a discontinuous first derivative would imply an infinite second derivative, and since the energy of the system is found using the second derivative, a discontinuous first derivative would imply an infinite energy, which again is not physically realistic.

# Examples



## 1. Particle Moving in 1-D $\psi(x)$

$$\hat{H}\psi = \hat{T}\psi + \hat{V}\psi = E\psi \longrightarrow -\frac{\hbar^2}{2m} \left( \frac{\partial^2 \psi}{\partial x^2} \right) + V(x)\psi = E\psi$$

- The form of  $V(x)$  depends on the physical situation:

- Free particle  $V(x) = 0$  for all  $x$ .
- Harmonic oscillator  $V(x) = \frac{1}{2}kx^2$

## 2. Particle Moving in 3-D $\psi(x,y,z)$

- SE  $\Rightarrow$ 

$$-\frac{\hbar^2}{2m} \left( \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} \right) + V(x, y, z)\psi = E\psi$$

or

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + V(x, y, z)\psi = E\psi$$

**Note:** The SE is a second order differential equation

# Solution of a free particle moving in 1-D



$$-\frac{\hbar^2}{2m} \left( \frac{\partial^2 \psi}{\partial x^2} \right) = E\psi$$

- This is a **second order differential equation** – with general solutions of the form:

$$\psi = A \sin kx + B \cos kx$$
$$\left( \frac{\partial^2 \psi}{\partial x^2} \right) = -k^2 (A \sin kx + B \cos kx) = -k^2 \psi$$

- SE  $\Rightarrow -\frac{\hbar^2}{2m} \left( \frac{\partial^2 \psi}{\partial x^2} \right) = \left( -\frac{\hbar^2}{2m} \right) \cdot (-k^2 \psi) = E\psi$

$$\Rightarrow E = \frac{k^2 \hbar^2}{2m} \quad (\text{i.e. } E \text{ depends on } k).$$

# Example: one-dimensional “particle in a box”

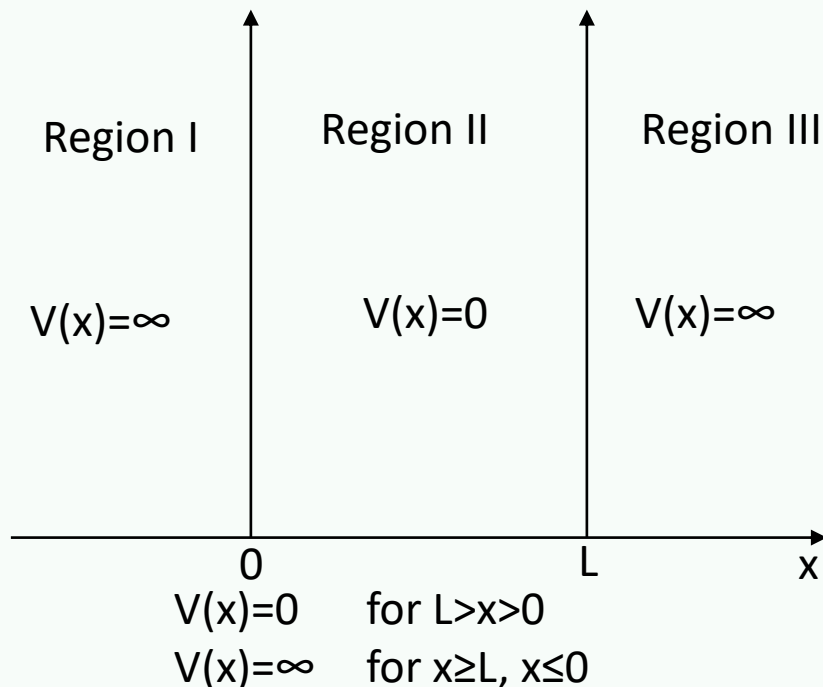
## System

Particle of mass  $m$  in 1-D box of length  $L$ .

Position  $x = 0 \rightarrow L$ .

Particle cannot escape from box as PE  $V(x) = \infty$  for  $x = 0, L$  (walls).

PE inside box:  $V(x) = 0$  for  $0 < x < L$ .



Time Independent Schrödinger Equation

$$\underbrace{\frac{-\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2}}_{\text{KE}} + \underbrace{V(x)\psi}_{\text{PE}} = \underbrace{E\psi}_{\text{TE}}$$

Applying boundary conditions:

Region I and III:

$$\frac{-\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + \infty * \psi = E\psi$$

Region II:

$$\frac{-\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E\psi$$

# contd.



## Restrictions on $\psi$

- In principle **Schrödinger Eqn.** has an infinite number of solutions.
- So far we have general solutions:
  - any value of  $\{A, B, k\} \rightarrow$  any value of  $\{\psi, E\}$ .
- **BUT** – due to the **Born interpretation of  $\psi$** , only certain values of  $\psi$  are physically acceptable:
  - outside box ( $x < 0, x > L$ )  $V = \infty \Rightarrow$  impossible for particle to be outside the box  
 $\therefore |\psi|^2 = 0 \Rightarrow \psi = 0$  outside box.
  - $\psi$  must be a continuous function  
 $\Rightarrow$  **Boundary Conditions**
 $\psi = 0$  at  $x = 0$   
 $\psi = 0$  at  $x = L$ .

# contd.



## Effect of Boundary Conditions

1.  $x = 0$

$$\psi = \underbrace{A \sin kx}_0 + \underbrace{B \cos kx}_1 = B$$

$$\psi = 0 \quad \therefore B = 0$$

$$\Rightarrow \psi = A \sin kx \quad \text{for all } x$$

2.  $x = L$

$$\psi = A \sin kL = 0$$

$A=0 ? \quad \times$   
(or  $\psi = 0$  for all  $x$ )

$\sin kL = 0 ? \quad \checkmark$

$$\sin kL = 0 \quad \Rightarrow \quad kL = n\pi \quad n = 1, 2, 3, \dots$$

( $n \neq 0$ , or  $\psi = 0$  for all  $x$ )

## Allowed Wavefunctions and Energies:

- $k$  is restricted to a discrete set of values:

$$k = n\pi/L$$

- Allowed wavefunctions:  $\psi_n = A \sin(n\pi x/L)$

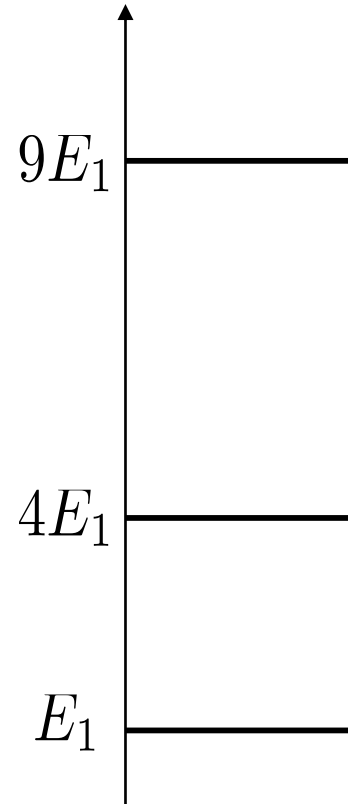
- Normalization:

$$A = \sqrt{2/L} \Rightarrow \psi_n = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

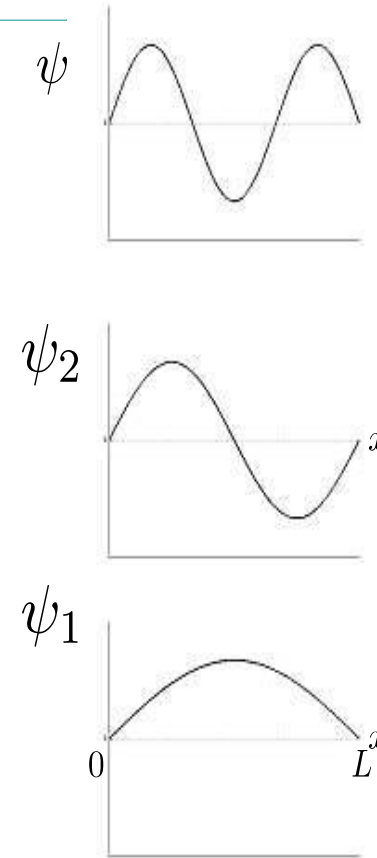
- Allowed energies:

$$E_n = \frac{k^2 \hbar^2}{2m} = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

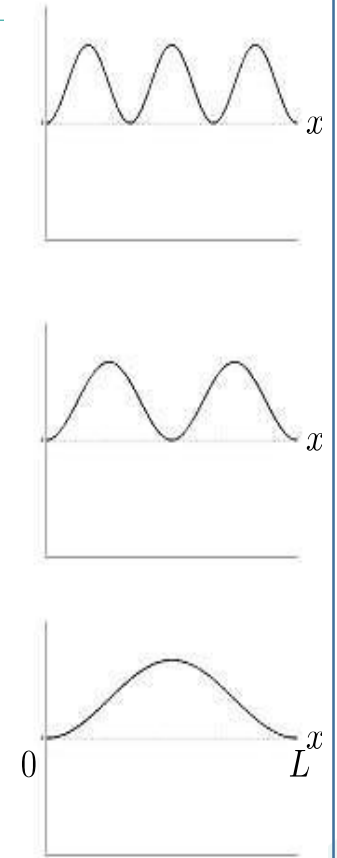
### EIGENENERGIES for 1-D BOX



### EIGENSTATES for 1-D BOX



### PROBABILITY DENSITIES



**Zero-Point Energy (ZPE)** – lowest energy of particle in box:

CM  $E_{\min} = 0$

QM  $E = 0$  corresponds to  $\psi = 0$  everywhere (forbidden).

# contd.

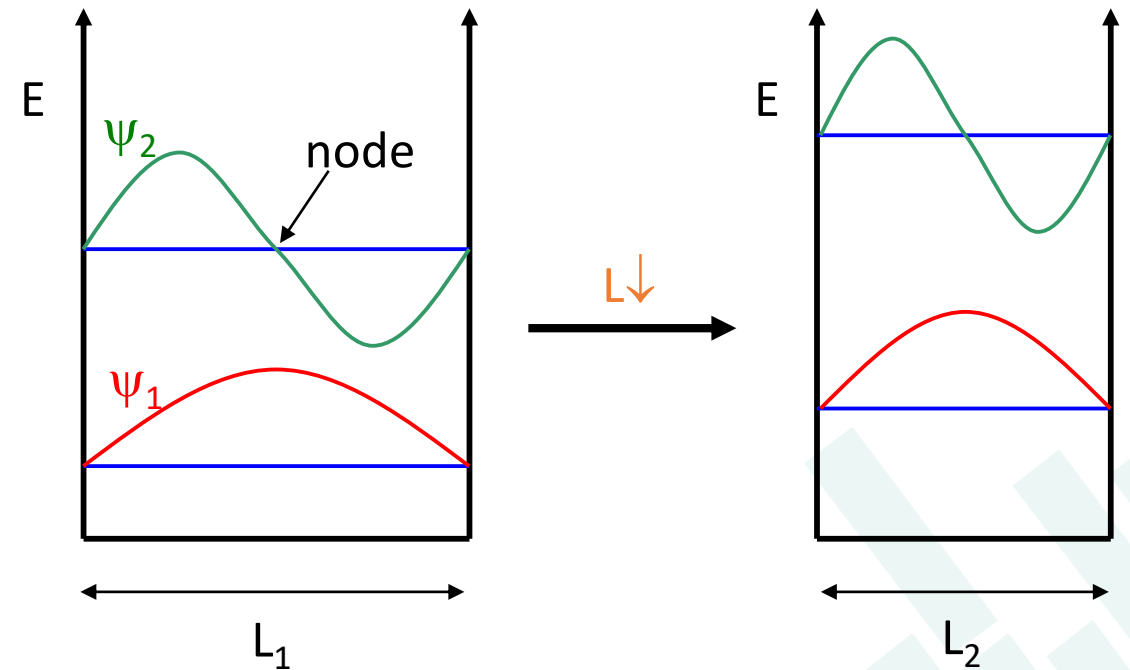


Energies

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

- $E_n \propto n^2/L^2 \Rightarrow E_n \uparrow$  as  $n \uparrow$  (more nodes in  $\psi_n$ )  
 $E_n \uparrow$  as  $L \downarrow$  (shorter box)

$n \uparrow$  (or  $L \downarrow$ )  $\Rightarrow$  curvature of  $\psi_n \uparrow$   
 $\Rightarrow$   $KE \uparrow \Rightarrow E_n \uparrow$



# Particle in a 2-Dimensional Box



A similar argument can be made:

$$\frac{-\hbar^2}{2m} \left( \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} \right) = E \psi$$

$$\psi(x, y) = X(x)Y(y)$$



Our Wave Equations:

$$\frac{-\hbar^2}{2m} \frac{\partial^2 X}{\partial x^2} = E_x \psi$$

$$\frac{-\hbar^2}{2m} \frac{\partial^2 Y}{\partial y^2} = E_y \psi$$

Doing the same thing to these differential equations that we did in one dimension we get:

$$X = \sqrt{\frac{2}{L_x}} \sin \frac{n_x \pi x}{L_x}$$

$$Y = \sqrt{\frac{2}{L_y}} \sin \frac{n_y \pi y}{L_y}$$

In one dimension we needed only one 'n'  
But in two dimensions we need an 'n' for the x and y component.

Since  $\psi = X(x)Y(y)$

$$\psi_{n_x n_y} = \sqrt{\frac{4}{L_x L_y}} \sin \frac{n_x \pi x}{L_x} \sin \frac{n_y \pi y}{L_y} \longrightarrow \psi_{n_x} = \sqrt{\frac{2}{L}} \sin \frac{n \pi x}{L}$$

For energy levels:

$$E_{n_x} = \frac{n^2 \hbar^2}{8mL^2} \longrightarrow E_{n_x n_y} = \frac{\hbar^2}{8m} \left[ \left( \frac{n_x}{L_x} \right)^2 + \left( \frac{n_y}{L_y} \right)^2 \right]$$

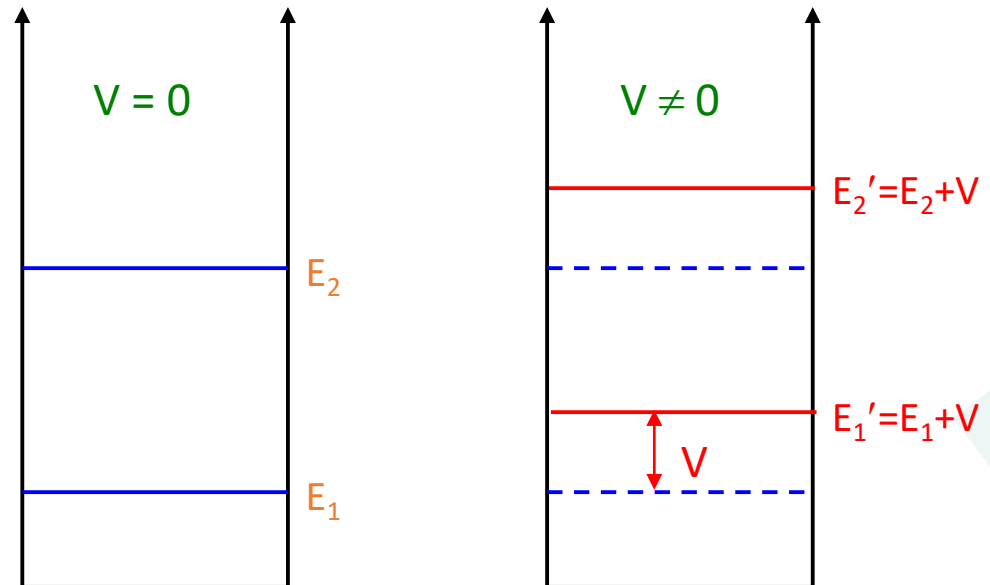
# If $V(x) = V \neq 0$



If  $V(x) = V \neq 0$ , everywhere in box

- If  $V(x) = V \neq 0$ , everywhere in box, all energies are shifted by  $V$ .

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} + V$$



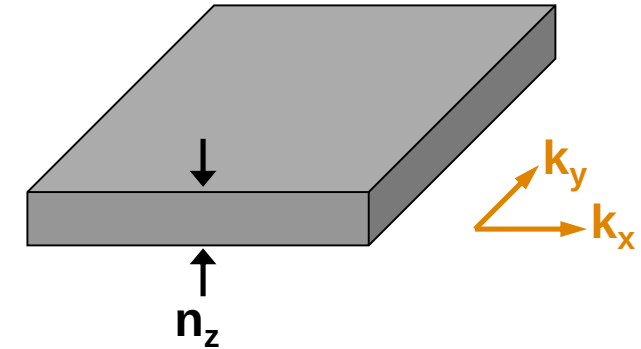
# Quantum Confinement in Nanostructures



Confined in:

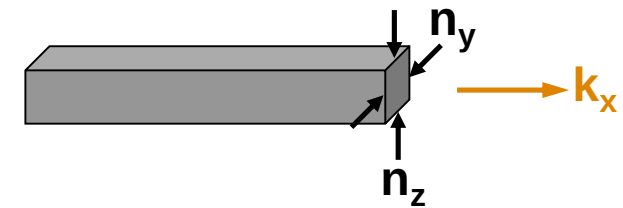
**1 Direction:** Quantum well (thin film)

Two-dimensional electrons



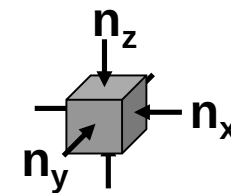
**2 Directions:** Quantum wire

One-dimensional electrons



**3 Directions:** Quantum dot

Zero-dimensional electrons



Each confinement direction converts a continuous  $k$  in a discrete quantum number  $n$ .